Environment Monitoring of Shanghai Nanhui Intertidal Zone With Dual-Polarimetric SAR Data Based on Deep Learning

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***Abstract*— Satellite-based synthetic aperture radar (SAR) can provide low-cost, frequent environment monitoring for dynamic intertidal zones. The critical problem is to realize pixel-level classification of SAR images of the intertidal zones with excellent and robust performance. Recently, deep learning, in particular deep convolutional neural networks, has provided us with promising solutions to this problem. Based on a sophisticated deep learning-based pixel-level classification model U^2-Net, we propose an MB-U^2-ACNet model suitable for intertidal zone land cover classification using dual-polarimetric SAR data integrated with environmental information, such as wind speed and tide level information. The MB-U^2-ACNet model has a multibranch nested U-shaped encoding–decoding structure. We extract and fuse features from multiple data sources, including satellite remote sensing and environmental information, by establishing the multibranch structure. Furthermore, we propose an asymmetric convolution residual U-block for each encoding–decoding stage to improve the model's feature extraction ability. Moreover, the model with attention mechanisms better distinguishes the importance of features from the channel's perspectives and spatial dimensions. We construct a dataset with 106 Sentinel-1 SAR images from 2016 to 2020 for environmental monitoring in the intertidal zone of Shanghai Nanhui. On the dataset, the proposed model reaches the overall classification accuracy of 96.40% and the mean intersection over union score of 0.8307. The experiments show the advantages of the proposed model compared with the benchmarking models due to better feature extraction and multisource information fusion. In addition, the contributions of every added substructure are analyzed systematically.**

***Index Terms*— Deep convolutional neural networks (DCNNs), deep learning, environment monitoring, intertidal zone, MB-U^2-ACNet, synthetic aperture radar (SAR) imagery.**

I. Introduction

BEING an essential part of the wetland, one of the three global ecosystems, the intertidal zone has a unique geographical location. Affected by the interaction between land and sea, the intertidal zone has various ecological service functions, such as coastal erosion protection, biological habitat support, water purification, and carbon sink. It plays a vital role in resisting the harm caused by storm surges and sea-level rise. Besides, the intertidal zone can bring many economic benefits in terms of land reclamation, mariculture, sea salt production, and port expansion [1]. At the same time, the intertidal zone is one of the most fragile ecosystems in the world. In the past 30 years, the area of tidal flats has decreased by 16.02% [2]. Extensive degradation from coastal development, sediment reduction in major rivers, river delta sinking, coastal erosion increasing, and sea-level rise, herald the continued negative trajectory of tidal flat ecosystems worldwide. As a highly dynamic area with essential environmental, economic, and social values, regular monitoring of the intertidal zone is essential.

For environment monitoring of the intertidal zone, a ground survey is a direct and accurate means, but it is costly and sparse sampling. Remote sensing is an economic and efficient means. Optical remote sensing is typically used [2], [3], [4], [5], [6], [7], [8], because the images are readily interpreted and provide multispectral sensing information. However, optical remote sensing is limited by lighting and weather conditions. The lack of optical remote sensing data in time and space is unfavorable for the rapid-changing intertidal zone monitoring. On the other hand, synthetic aperture radar (SAR) is an active

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microwave sensor. Its image quality is less affected by weather and light conditions. Therefore, SAR has become an essential means to monitor environmental changes in the intertidal zone [5], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19]. However, SAR images are difficult for end-users to understand. Therefore, accurate and efficient information extraction methods are required. In recent years, studies pointed out the challenges in analyzing coastal environments, such as wetlands, mangroves, mudflats, and intertidal areas, with polarimetric C-band SAR data like this research. Geng *et al.* [20] studied the polarimetric characteristics of mud flats in an intertidal zone using C-band and X-band spaceborne SAR data. Muro *et al.* [21] performed short-term change detection in wetlands using Sentinel-1 time-series. Ferrentino *et al.* [22] researched on the use of multipolarimetric C-band SAR data for coastline extraction in harsh coastal environments. Hu *et al.* [23] presented the mapping of coastal salt marshes in China using the time series of Sentinel-1 SAR.

The essential problem for SAR image-based environment monitoring of the intertidal zone is accurate pixel-level land cover classification over this zone, which is the research topic of this article. Methods along different research directions have been proposed for pixel-level classification of SAR images. These methods can be categorized into three types: 1) the methods based on SAR scattering characteristics [24], [25], [26]; 2) the methods based on statistical analysis [27], [28], [29], [30]; and 3) the methods based on feature extraction and machine learning, such as the support vector machine (SVM) [31], the random forest [32], and the artificial neural network [33], [34], [35], [36]. In addition, some studies combined the ideas from different categories [37], [38]. For SAR image classification, the integration of contextual information is also a research focus [39], [40], [41], [42], [43], [44]. The aforementioned methods are theoretically sound, achieved good classification performances, and solved practical problems. However, these methods are based on human-crafted features and rules from analyzing a small number of samples. Therefore, it is difficult for them to provide robust performance under various influential factors when facing an era of big SAR data.

The deep learning technology, especially the deep convolutional neural network (DCNN) models, provides a promising direction for robust SAR image classification. In the DCNN models, the features for robust SAR image pattern classification are mined from multidimensional SAR data instead of being predefined. These data-driven models can provide robust features under various influential factors, and they are optimized in an end-to-end way, from data to information. Recent studies [45], [46], [47], [48] have verified this idea in the geoscience and remote sensing community.

Recently, DCNN-based methods have already demonstrated their advantages in SAR image classification. The first batch of studies converted the pixelwise classification of SAR images into the patch-level classification solved by DCNN models. Zhou *et al.* [49] explored the DCNN-based model for polarimetric SAR image classification and achieved comparable performance with state-of-the-art. Zhang *et al.* [50] extended CNN to a complex value domain to use polarization phase information. Chen and Tao [51] tried to incorporate the expert knowledge of target scattering mechanism interpretation and polarimetric feature mining to assist the DCNN model training and improve the classification performance. Qin *et al.* [52] proposed a superpixel-oriented complex-valued DCNN model for incorporating contextual information explicitly. The DCNN models have shown their potential for robust SAR image classification in the last several years. With the help of the target scattering mechanism information, the DCNN models can be improved.

With the development of fully convolutional neural networks (FCNNs), a type of the DCNN model, FCNN-based models have gradually been used for pixelwise classification of SAR images. The FCNN models label the image on a pixel-to-pixel basis, unlike the above-mentioned models. Henry *et al.* [53] evaluated different FCNN models and demonstrated that the FCNN models can provide an effective solution for road extraction in SAR images. Liu *et al.* [46] presented a specially tailored FCNN model for efficient and robust coastal inundation mapping from bitemporal and dual-polarization SAR images with the help of radar scattering mechanisms. Mullissa *et al.* [54] presented an FCNN model that uses real-valued weight kernels to perform pixelwise classification of complex-valued images. They found that the proposed method achieved higher accuracy compared with all other compared networks with the same architecture. Zhang *et al.* [55] developed a U-shaped FCNN model taking full-scale skip connections for waterline extraction in Subei Sandbanks, China, from SAR images. Detailed research on the development of the DCNN-based methods for SAR image classification can be found in the recent review papers [56], [57].

Although the above-mentioned methods show superior classification performance by continuously improving the models in classification tasks, in essence, the model development does not focus on multichannel information fusion for SAR image classification. Observations with multiple parameters, such as multiple polarizations, multiple frequencies, multiple incidence angles, and multiple imaging dates, can get richer information about the scene. In addition, the observation of the intertidal zone is influenced by the physical environment factors. The integration of physical environment factors may improve classification performance. Therefore, information fusion in the DCNN framework may improve the model's classification performances and generalization abilities. Although the aforementioned DCNN-based methods often combine data from multiple channels, they merge data into multichannel image cubes, sometimes with some transformation, such as pseudocolorization [58], and use the cubes for model training and application. In the developed models, the network structure for different information channels is shared; thus, the information fusion may not be efficient or adaptive.

Researchers of the remote sensing community try to design model structures constructed from multiple branches to use data from different sources adaptively. Xu *et al.* [59] proposed the dual-branch method to use hyperspectral imaging and light detection and ranging (LiDAR) data as input information to enhance the extraction of corresponding spectral–spatial

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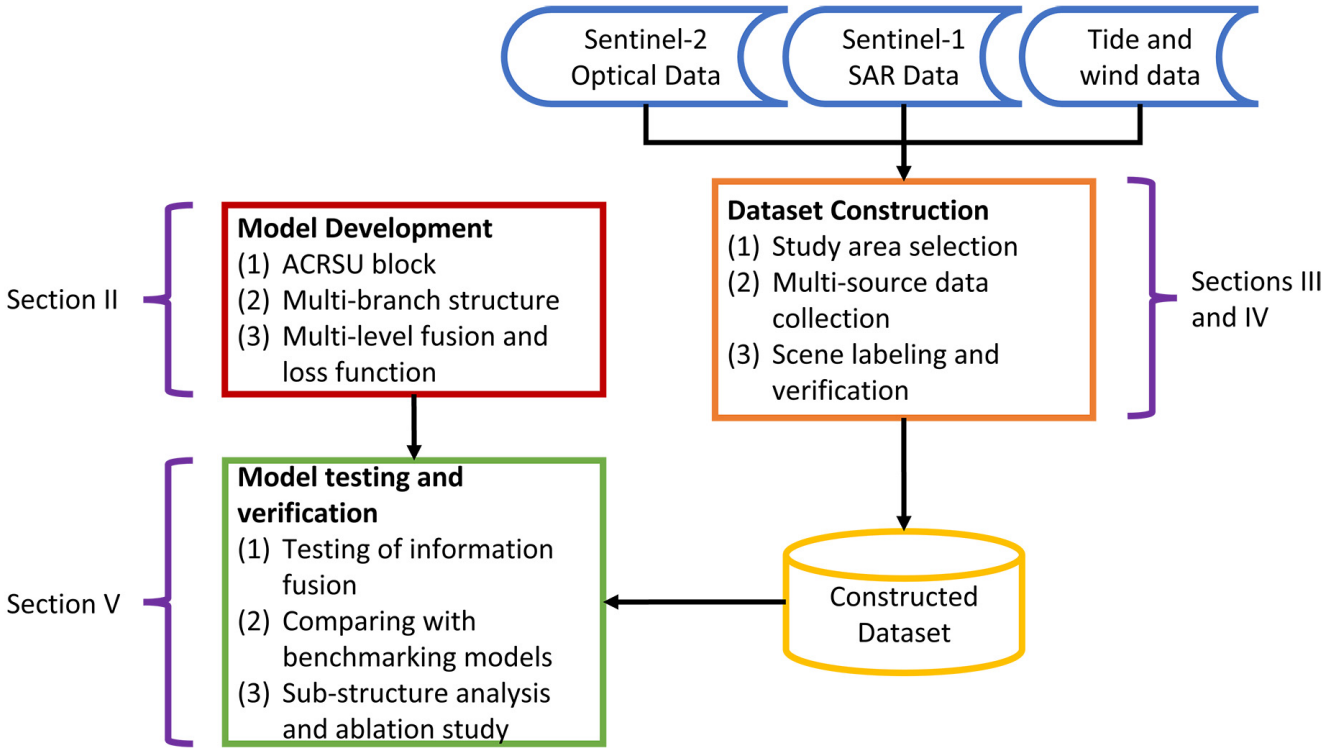


Fig. 1. Illustration of the structure of this study, consisting of model development, dataset construction, and model testing and verification.

feature information. Based on the work of Zhou *et al.* [49], Gao *et al.* [60] added a PauliRGB branch to extract images' spatial features. Feng *et al.* [61] developed a multibranch DCNN model to combine Sentinel-1 and Sentinel-2 data to classify coastal land covers. Qiu *et al.* [62] fused Sentinel-2 images of four seasons, which proved the potential of multiseason images in urban land cover classification. Although the above-mentioned methods all use a branching structure to extract different feature information, the following problems still exist: 1) the existing network structures are relatively simple and cannot fully extract the image's high-level contextual feature information. 2) The fusion of features from different network branches often happens in the late part of the network. The fusion of feature information generated at different stages deserves exploration. 3) The multibranch structure for the fusion of multipolarization information and physical environment factors should be developed for robust multichannel SAR image classification in intertidal zone environment monitoring use.

Considering the above-mentioned aspects deserving exploration, we propose a multibranch DCNN model for fusing dual-polarimetric SAR and physical environment information, such as wind speed and tide level information, for land cover classification of the intertidal zone. The model was tailored from the U^2-Net framework [63] with asymmetric convolutions. We name it the MB-U^2-ACNet model [64]. It has several characteristics.

1) The MB-U^2-ACNet model has a multibranch nested U-shaped encoding–decoding structure. We extract and fuse features from multiple data sources, including satellite remote sensing and environmental information, by establishing the multibranch structure. By means of the model, we can demonstrate the positive contributions of the integration of dual-polarization and environmental factors in intertidal zone SAR image classification.
2) We design an asymmetric convolution residual U-block (ACRSU) for each encoding–decoding stage to improve the model's robustness.
3) The model with attention mechanisms better distinguishes the importance of features from the channel's perspectives and spatial dimensions.

The structure of this study is illustrated in Fig. 1, consisting of model development, dataset construction, and model testing and verification. The descriptions of these parts are organized as follows. Section II introduces the proposed network model in detail, including the overall structure, branching structure, and ACRSU. Sections III and IV describe the study area and dataset dedicated to Shanghai Nanhui intertidal zone environment monitoring. In Section V, performance evaluation and comparison are presented in detail, and corresponding analysis and discussions are given. Finally, Section VI gives the conclusions.

II. Methodology

This section introduces the proposed network, MB-U^2-ACNet, in detail. This section is organized as follows. First, the basic block for multiscale feature extraction and the design for multibranch feature fusion are described in detail. Then, the network is presented to show the functions of its main parts. Finally, the loss function used in the network to achieve its goal is introduced.

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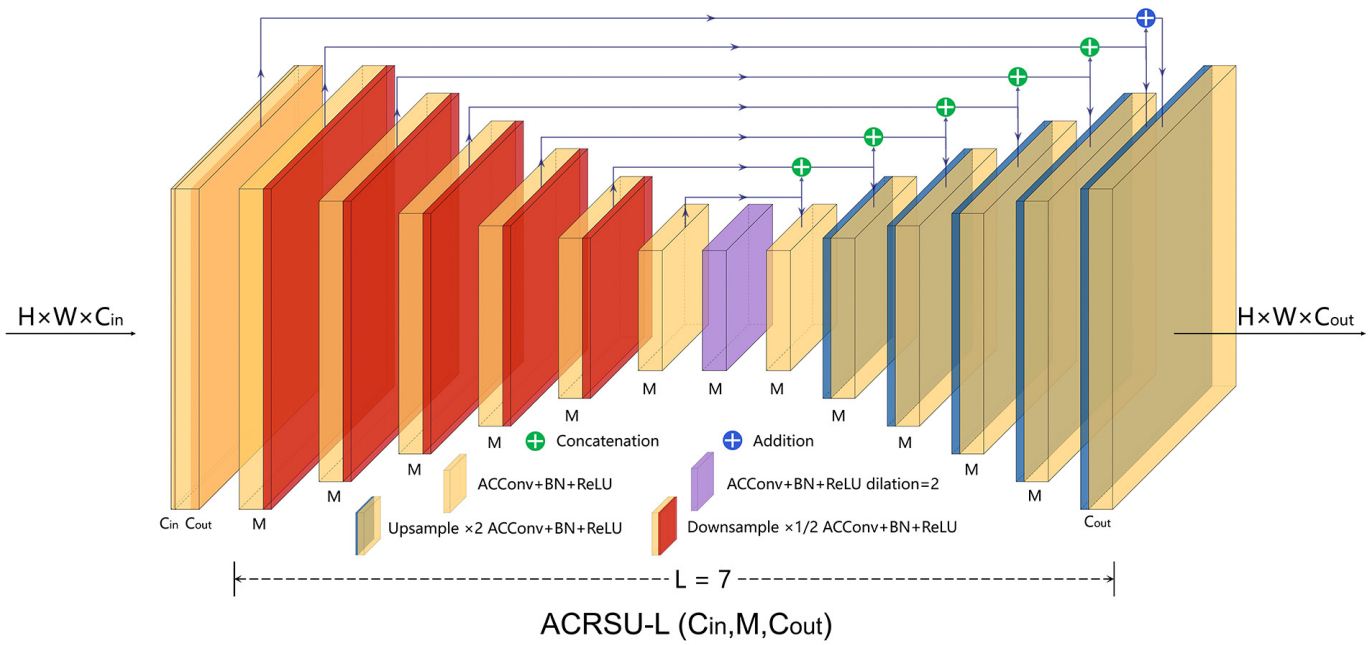


Fig. 2. Designed ACRSU. L is the encoder's depth. C_{in} and C_{out} represent the channel numbers of the input and output feature maps, respectively. M is the channel number of the middle layers.

A. ACRSU Block

The RSU is a functional block used for multiscale feature extraction in U^2-Net [63], filled in each stage of the encoder–decoder structure. Its interior design is a symmetrical structure similar to U-Net, composed of several convolution, pooling, and batch normalization [65] operation layers with different sizes. These operation layers can implement nonlinear operations on input data to achieve the purpose of feature extraction. RSU increases the downsampling operation process compared with traditional convolutions, thereby achieving more delicate feature extraction with less computational overhead.

This study applies the RSU block initially designed for computer vision tasks to the SAR image classification task. To improve RSU's ability to extract local and global contextual feature information and improve SAR image classification and recognition capabilities at different flip and rotation angles. Inspired by the asymmetric convolution [66], we propose the ACRSU.

The ACRSU-L (C_{in}, M, and C_{out}) shown in Fig. 2 follows the main structure of the RSU in U^2-Net. L denotes the depth of the ACRSU block. C_{in}, M, and C_{out} denote the channel numbers of the input, middle, and output feature maps, respectively. Thus, the ACRSU block mainly consists of the following three parts.

1) For local feature extraction, it uses the asymmetric convolution instead of the vanilla convolution in the original RSU block. Compared with the RSU, the ACRSU using asymmetric convolution does not require additional parameters and calculations during the inference process while enhancing the skeleton part of the square convolution kernels and improving the accuracy. This layer converts the input feature map x ($H \times W \times C_{\text{in}}$, where H and W denote height and width, respectively) into an intermediate image $F(x)$ with C_{out} channels.
2) A U-Net [67] like symmetric encoder–decoder structure with height L is used. A larger L leads to a deeper ACRSU, more convolution operations, and richer feature extraction. It takes $F(x)$ of intermediate feature map as input and learns to extract and encode multiscale context information $U(F(x))$, where $U(\cdot)$ represents a structure similar to U-Net, as shown in Fig. 2.
3) A residual connection that merges local features and multiscale features through the summation of $F(x)$ and $U(F(x))$.

ACRSU has the same U-like structure as RSU, and it can realize the function of multiscale feature extraction. The ACRSU blocks are the basic components of the proposed MB-U^2-ACNet model and they are used as encoders and decoders in the model. The main difference between ACRSU and RSU is that ACRSU replaces the $d \times d$ square convolution kernels in RSU with an asymmetric convolution with three branches and sizes of $d \times d$, $1 \times d$, and $d \times 1$, as shown in Fig. 3. We suppose that several 2-D kernels with compatible sizes operate on the same input with the same stride to produce the same resolution outputs, and their outputs are summed up. Then, we can add up these kernels in the corresponding positions. According to the convolution property, the corresponding positions are added to obtain an equivalent kernel that produces the same output as the summed-up output. The process can be expressed as the following equation. The detailed information of the asymmetric convolution can be found in [66]

$$I \otimes K^{(1)} + I \otimes K^{(2)} = I \otimes \left(K^{(1)} \oplus K^{(2)}\right) \tag{1}$$

where I is a matrix; $K^{(1)}$ and $K^{(2)}$ are size-compatible convolution kernels; $\otimes$ denotes the convolution process; and

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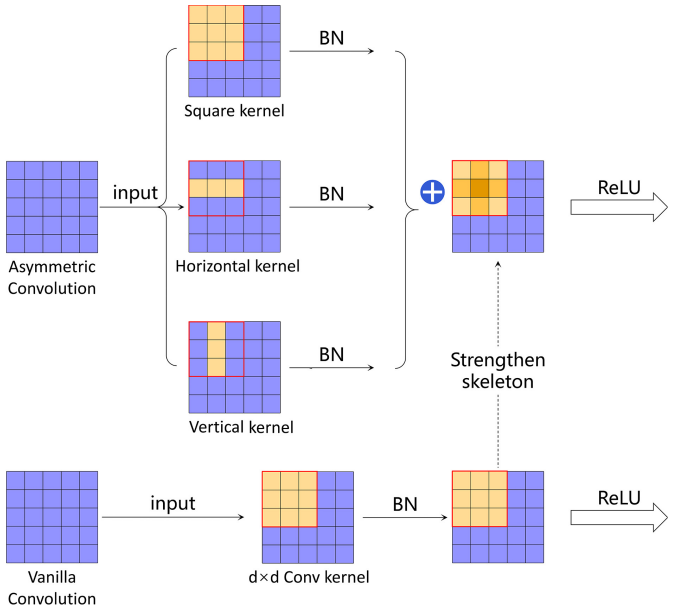


Fig. 3. Comparison of the asymmetric convolution and the vanilla convolution. In this figure, the asymmetric convolution contains three convolutional layers with kernel sizes of 3×3, 1×3, and 3×1, and they use the same output. Adapted from [66].

$\oplus$ denotes the addition of the parameters of the convolution kernels at the corresponding positions.

Without increasing the amount of parameter calculation, because of the addition of the horizontal and vertical convolutions, the fusion of the three convolution branches enhances the skeleton's feature extraction ability of the convolution kernel. Besides, trained asymmetric convolutions can still extract correct features from inverted images, making ACRSU more efficient in interpreting images under the conditions of inversion and rotation. Enhancing local feature extraction capabilities and enhancement of image interpretation capabilities under inversion and rotation conditions can be effective in improving the performance and generalization ability of the SAR image classification model.

B. Structure of Multibranch Feature Fusion

In the multipolarization SAR observation system, SAR images from different polarizations may have different representations of the same land cover type, different intensities, textures, and/or structures. For the SAR image classification task, the contributions of the images from different polarization channels to recognizing land cover types may be different. Therefore, for the SAR image classification task, a multibranch structure is often constructed to improve performance. At present, most network models with branching structures are designed as unidirectional DCNN structures, and feature fusion operations are placed at the end of the structure. This design will result in a lack of information interaction between different branch units, and the information after feature fusion cannot fully represent the characteristics of different branch units. In the information transmission process of each branch unit, to be able to give full play to the information sharing between different branch transmission data, we combined the encoder–decoder structure and designed the multibranch feature fusion structure, as shown in Fig. 4 and Table I.

In this study, we focus on Sentinel-1 dual-polarization SAR data. Therefore, the model is a dual-branch structure for dual-polarization information fusion. Besides, wind speed and tide level information are influential physical environment factors. We suppose that the inclusion of this information positively impacts identifying SAR data and the surface environment. Thus, we also combine wind speed and tide level information into a polarization branch in a cascade manner.

The branching structure in this study is a symmetrical structure with encoding and decoding parts. The encoders are composed of different branch units, and the operation in each branch unit is the same. The encoders of the first five levels are denoted as En_p_q, where $p \in \{1, 2, 3, 4, 5\}$ denotes that the encoder is the pth-level encoder in the network structure, and $q \in \{1, 2\}$ denotes that the encoder is in the qth branch. The encoder of the sixth level is denoted as En_6. The decoders of the first five levels are denoted as De_p, where $p \in \{1, 2, 3, 4, 5\}$ denotes that the decoder is the pth-level decoder in the network structure. The ACRSU blocks in the first four levels of encoders and decoders use the same convolution and pooling downsampling operations. The depths of the encoders/decoders gradually decrease from 7 in the first level to 4 in the fourth level. The depth L of En_5_q, En_6, and De_5 is consistent with En_4_q. For encoders and decoders in the higher levels, En_5_q, En_6, and De_5, to avoid information loss caused by downsampling, ACRSU-4F blocks use dilated convolutions instead of pooling operations. In En_5_q, En_6, and De_5, the resolution of the intermediate feature maps is the same as the resolution of the input and output feature maps. The ACRSU-4F block is illustrated in Fig. 4.

First, when the information from different data sources is passed into the multibranch feature fusion structure, it will be entered into different branch units. This information will be encoded by an encoder composed of an ACRSU block with depth L in each stage. Output feature maps are then passed into spatial attention modules [65] to capture feature maps' differentiated feature information by respective branch units. Finally, the feature maps generated by spatial attention modules will be concatenated together to the same-level decoders. After the above operations, the branching structure transfers the encoder's information to the decoder of the same level. It avoids the loss of local and global feature information in the transmission process caused by dimensionality reduction and direct addition.

C. MB-U^2-ACNet

This study proposed a multibranch DCNN model tailored explicitly from the U^2-Net framework and with asymmetric convolutions, which we named MB-U^2-ACNet. Fig. 5 illustrates the proposed MB-U^2-ACNet model. The network structure can be divided into four parts.

1) It is a multibranch structure for feature fusion. At the input end, according to different data sources, different branches are selected as input. At different levels of

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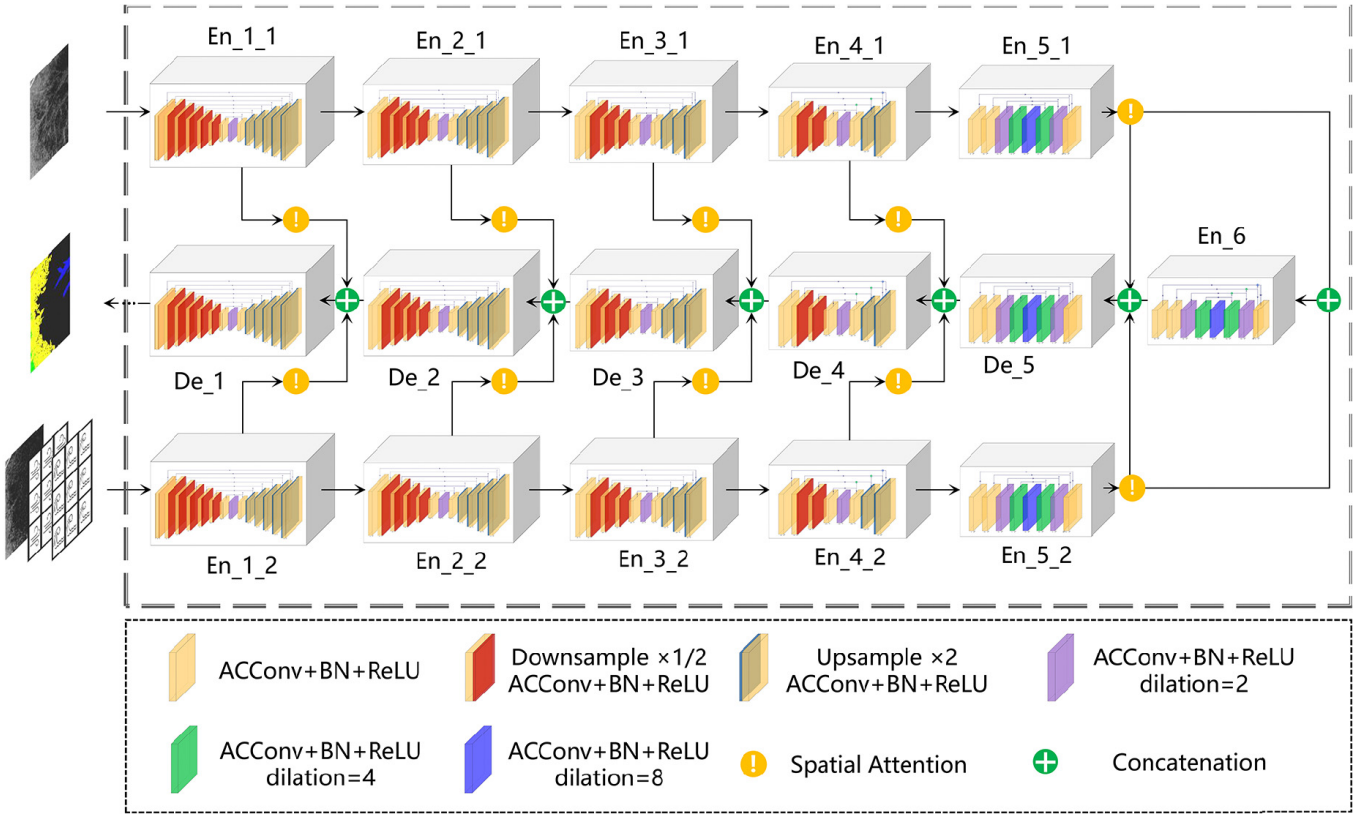


Fig. 4. Illustration of the multibranch structure. Branch 1 deals with the VV polarization channel data, and branch 2 deals with the VH polarization channel data integrated with wind speed and tide level information.

TABLE I
DETAILED CONFIGURATION OF THE ACRSU BLOCK AT EACH STAGE

	En_1_1	En_2_1	En_3_1	En_4_1	En_5_1	En_6
Block	ACRSU-7	ACRSU-6	ACRSU-5	ACRSU-4	ACRSU-4F	ACRSU-4F
C_{in}	1	64	128	256	512	1024
M	32	32	64	128	256	256
C_{out}	64	128	256	512	512	512
	En_1_2	En_2_2	En_3_2	En_4_2	En_5_2	
Block	ACRSU-7	ACRSU-6	ACRSU-5	ACRSU-4	ACRSU-4F	
C_{in}	3	64	128	256	512	
M	32	32	64	128	256	
C_{out}	64	128	256	512	512	
	De_5	De_4	De_3	De_2	De_1	
Block	ACRSU-4F	ACRSU-4	ACRSU-5	ACRSU-6	ACRSU-7	
C_{in}	1536	1536	1024	512	256	
M	256	256	128	64	16	
C_{out}	512	512	256	128	64	

the branching structure, the intermediate outputs will be connected to spatial attention modules, put together in a concatenation manner, and then connected to the decoders of the same levels.

2) It has a six-level encoding–decoding structure. Detailed configurations are listed in Table I. Different branch units have the same encoding configuration, and features generated from the encoder will go in two directions. In one direction, the feature maps are directly passed to the next-level encoder. In the other direction, the feature maps are put into the multibranch feature fusion after the spatial attention modules.
3) The five-level decoders and the six-level encoders have a similar and basic design with different configuration parameters (see Fig. 2 and Table I). At each level, the decoder's input comes from two parts. One part is the output of the decoder of the previous level after upsampling. The other part is the fusion output of the same-level encoders of different branches after spatial attention modules.
4) The last part is the channel attention mechanism of the decoding sides. At each level, the input of the decoder comes from different information branches, making feature differentiation between channels more

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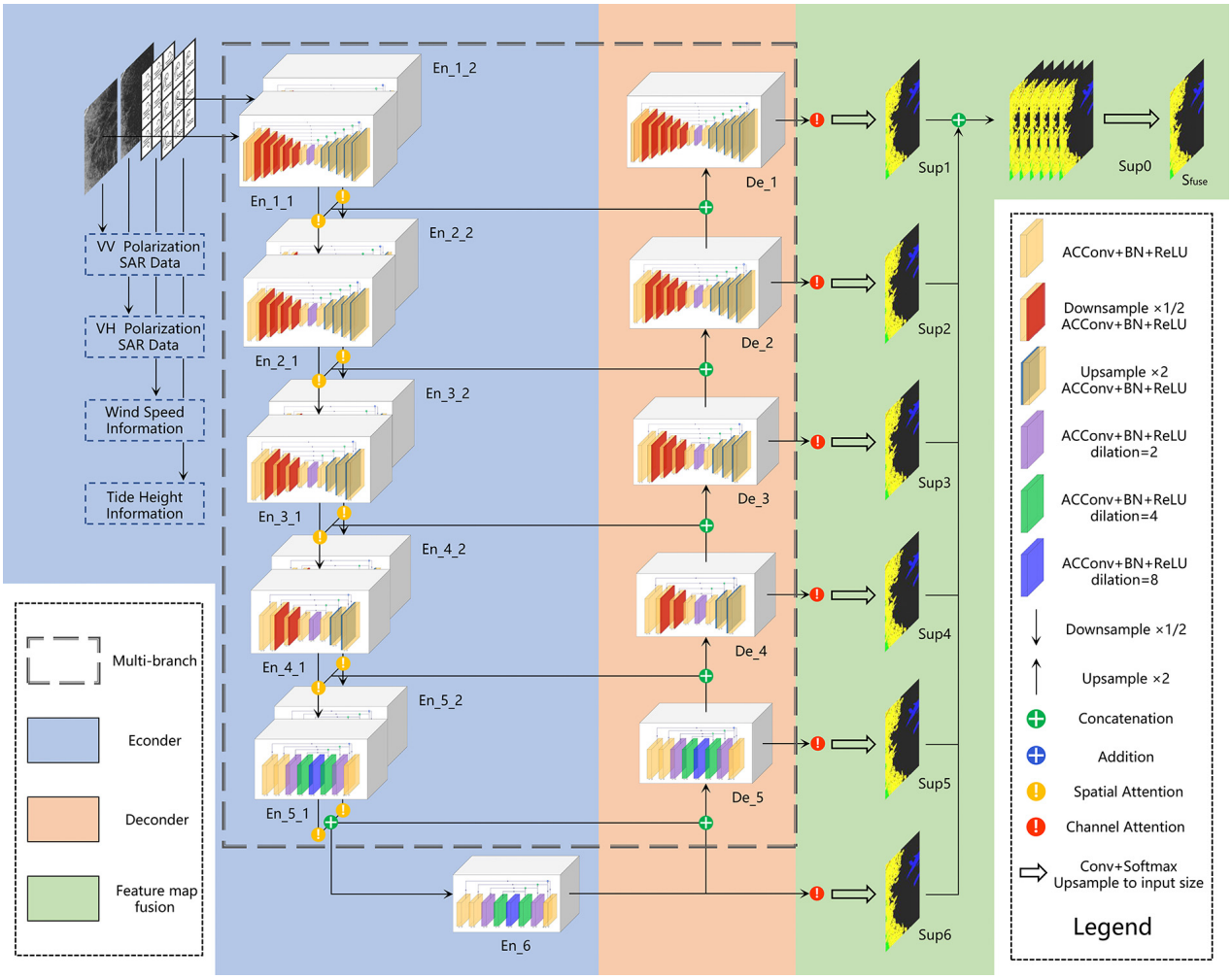


Fig. 5. Schematic of the MB-U^2-ACNet model. Each of these stages consists of the proposed ACRSU.

obvious. Therefore, we connect the channel attention module [65] after the feature maps outputted by the decoder side of each level to realize the redistribution of the channels' weights of the feature maps.

D. Multilevel Fusion and Loss Function

In multiclass classification tasks, the most commonly used loss function is the cross-entropy (CE) loss. For this study, we design the loss from the original CE loss. Via the visual observation of labeled images, we found an obvious imbalance in the sample numbers of classes, making predictions tend to the classes with more sample numbers. Therefore, the loss function will weigh in sample numbers to balance this trend. The function of loss calculation for each pixel is

$$l_{\mathrm{WCE}} = -\sum_{i=1}^{N} \lambda_i t_i \log\left(\frac{\exp(s_i)}{\sum_{i=1}^{N}\exp(s_i)}\right) \tag{2}$$

where N denotes the number of categories. λ_i is the weight for the ith category in loss calculation, and it is defined as the multiplicative inverse of the sample number proportion of the ith category. t_i indicates the ground truth. If the ground truth of this pixel is the ith category, $t_i = 1$; otherwise, $t_i = 0$. The calculation inside $\log(\cdot)$ uses the model's output s_i to predict the probability of this pixel belonging to the ith category. From the pixel loss l_{WCE} to the batch loss L_{WCE}, we just have to add losses of all the pixels in the batch.

Inspired by [63], our loss function is defined as

$$L = w_{\mathrm{WCE}}^{(\mathrm{Sup0})} L_{\mathrm{WCE}}^{(\mathrm{Sup0})} + w_{\mathrm{WCE}}^{(\mathrm{Sup1})} L_{\mathrm{WCE}}^{(\mathrm{Sup1})} + \cdots + w_{\mathrm{WCE}}^{(\mathrm{Sup6})} L_{\mathrm{WCE}}^{(\mathrm{Sup6})} \tag{3}$$

where $L_{\mathrm{WCE}}^{(\mathrm{Sup1})}, L_{\mathrm{WCE}}^{(\mathrm{Sup2})}, \ldots, L_{\mathrm{WCE}}^{(\mathrm{Sup6})}$ are the losses calculated from side predictions with ground truth, and $L_{\mathrm{WCE}}^{(\mathrm{Sup0})}$ is the loss calculated from fused predictions with ground truth, as illustrated by Fig. 5. $w_{\mathrm{WCE}}^{(\cdot)}$ denotes corresponding weights.

III. Study Area

Nanhui intertidal zone in Shanghai, China, extends from the airport enclosure area in the north to the Luchao Port in the south. It is one of four major intertidal zones in Shanghai, accounting for about 11% of the total area of Shanghai intertidal zones. It is located at the Yangtze River Estuary and Hangzhou Bay intersection. The large amount of silt carried by the upstream is silted up here, promoting the intertidal zone to extend to the sea continuously. Due to the reclamation project and the strong tidal current, the intertidal zone is distributed with vegetation, sediment, and other land covers. Since about ten years ago, this intertidal zone has been a planned area for

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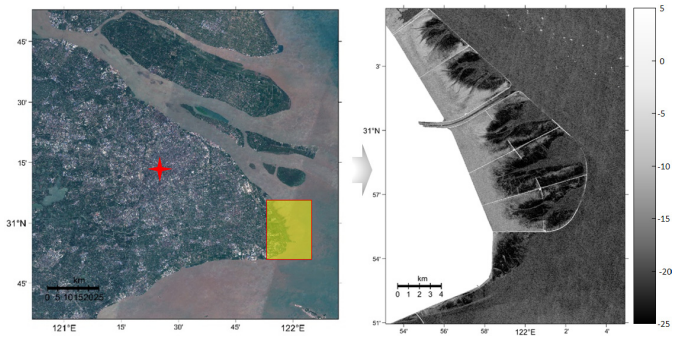


Fig. 6. Geographical location of the study area, the intertidal zone of Nanhui, Shanghai. The left is Sentinel-2 optical composited image. The right subimage is the Sentinel-1 VV polarization channel image on September 15, 2019. The masked-out area in white is not considered in the model evaluation.

promoting the sedimentation of Shanghai Nanhui. As shown in Fig. 6, the study area is located on the south side of the remediation area of the first-stage sedimentation promotion project N1 and the east side of Donghai Bridge. The geographical position coordinates are 30°50′56″–31°5′42″N and 121°53′7″–122°4′56″E. The total area is about 600 km^2. In Fig. 6, the left part is Sentinel-2 optical composited image. The right subimage is the Sentinel-1 vertical transmit and vertical receive (VV) polarization channel image on September 15, 2019. The masked-out area in white is not considered in the model evaluation.

IV. Data

In this study, the primary source of information is Sentinel-1 [68] SAR data with VV and vertical transmit and horizontal receive (VH) polarization channels. Sentinel-2 [69] optical data are used as auxiliary information to mark vegetation areas in the intertidal zone of the study area. Sentinel-1 is equipped with C-band SAR sensors, achieving single-polarization and dual-polarization sensing means. Sentinel-1A was launched on April 3, 2014, and the revisit period was 12 days. Sentinel-1B was launched on April 25, 2016. After the successful launch of Sentinel-1B, the revisit cycle was shortened from 12 days to 6 days via networking. Sentinel-2 carries multispectral imagers, covering 13 spectral bands with 10–60-m spatial resolution and a revisit period of ten days. Therefore, the revisit period can be up to five days after the two satellites are networked. Due to the long revisit time interval of Sentinel-1 in 2015, the study period is from January 2016 to October 2020. The data are downloaded from the Google Earth Engine (GEE) [70] platform.

A. SAR Data

SAR data used in this study are the level-1 ground range detected (GRD) dual-polarization (VV/VH) data of Sentinel-1. The imaging mode is the interferometric wide swath, and the pixel spacing is about 10 m × 10 m. It is worth noting that the VV polarization data are sensitive to soil moisture and can distinguish flooded from nonflooded vegetation. Likewise, the VH polarization data can help monitor herbaceous wetland vegetation [71]. The cross-polarization information is generated by the volumetric scattering in the vegetation canopy and has a higher sensitivity to the vegetation structure.

The GEE platform performs preprocessing on the Sentinel-1 data, including updating orbital metadata, removing GRD boundary noise, removing thermal noise, and performing radiometric calibration and terrain correction. These steps generate geocoded backscattering intensity images. Then, the backscattering intensity images are converted into log-scale normalized backscattering coefficient values (σ^0_{dB}). Finally, we perform linear enhancement for better visualization to facilitate image annotation. The linear enhancement is performed automatically. For the VV polarization, the upper and lower bounds are 5 and −25, respectively; for the VH, 0 and −30, respectively.

B. Optical Data

In this study, we use Sentinel-2 optical data as auxiliary data. Sentinel-2 has 13 spectral bands at different spatial resolutions, including four bands at 10-m: blue, green, red, and near-infrared, six at 20-m, and three with 60-m spatial resolutions. In this study, the vegetation normalization index (NDVI) is used as auxiliary information to divide the vegetation areas into dense and sparse vegetation types. The NDVI index [72] is one of the most commonly used vegetation indices to characterize seasonal and interannual changes.

The NDVI index is sensitive to photosynthesis biomass and divides areas with different vegetation coverage via thresholds. According to the characteristics of this study area, the threshold of NDVI is set to 0.26, and the areas with NDVI values greater than 0.26 are characterized as high-density vegetation areas. The extraction of NDVI is affected by the cloud coverage of the collected optical image. The QA60 bitmask band provided by GEE can detect and mask clouds and cirrus clouds. Furthermore, the revisit period of the Sentinel-2 dual-satellite network is relatively short; so we filter all Sentinel-2 images within 14 days with cloud coverage less than 20%, and the median values of these images are used to calculate the NDVI index values.

C. Tide Level and Wind Speed Data

Tide level and wind speed are used as environmental factors and integrated with Sentinel-1 SAR data in the proposed model to improve its performance. The tide level information comes from the tide level of the Shanghai Zhongjun Station on the tide table [73], [74], [75], [76], [77], and its geographical coordinates are 31°07′N and 121°54′E. The nearby wind speed information in our study is provided by the National Oceanic and Atmospheric Administration, Washington, DC, USA [78] at the time of the Sentinel-1 data acquisitions.

D. Scene Labeling and Verification

Using the VV and VH SAR backscattering information as the primary information, with the help of optical images, the scene is manually labeled into vegetation, sediment, and water. Sediment and water are divided mainly according to the VV channel information, and vegetation and nonvegetation areas are divided mainly according to the VH channel information. With the help of the NDVI values, the vegetation areas are further labeled into dense and sparse vegetation areas. The

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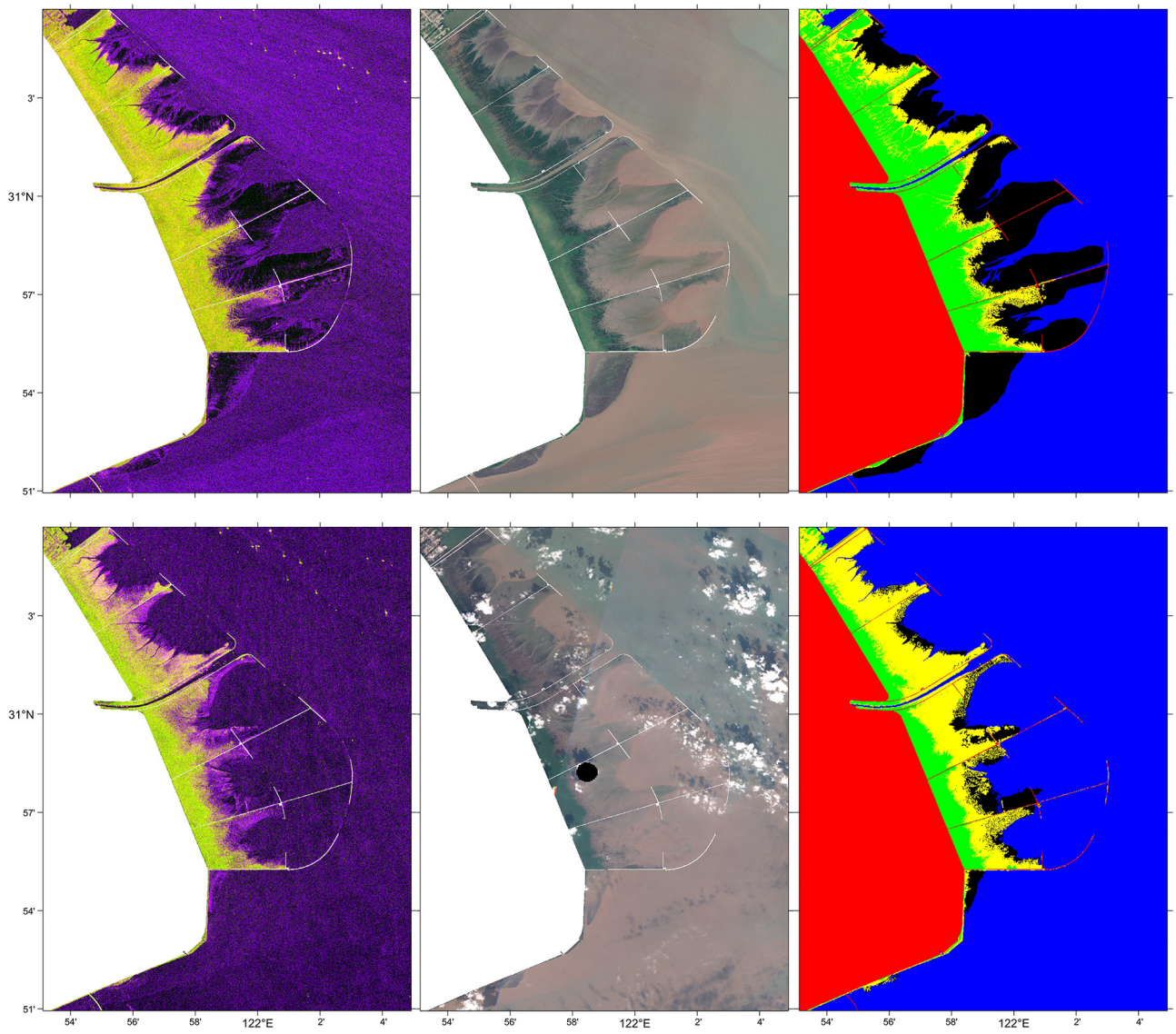


Fig. 7. Two images selected from the dataset. First row: August 22, 2020 (Sentinel-2 optical image in mid-August, 2020). Second row: November 8, 2019 (Sentinel-2 optical image in the first half month of November, 2019). Left: Sentinel-1 dual-polarization pseudocolor RGB SAR image [R: VV (dB), G: VH (dB), and B: VV − VH (dB)]. Middle: Sentinel-2 optical composited image around Sentinel-1 sensing time. Right: Labeled ground truth. Green, yellow, black, and blue indicates dense vegetation, sparse vegetation, sediment, and water, respectively. In the left and middle columns, the white area indicates the unstudied area. In the right column, the red area indicates the unstudied area.

inland areas are considered as background, and they are not used in performance evaluation. Considering the wind speed information and the cloud coverage information of the Sentinel-2 images within 14 days, we can finally screen out a total of 106 images from January 16, 2016, to October 3, 2020, and then label them. Fig. 7 shows two images with labels of August 22, 2020, and November 8, 2019.

On September 10, 2020, we conducted a field survey of the study area, collecting 76 photographs by a camera on a drone. Their global positioning system (GPS) coordinates were recorded, and eight photographs were selected for comparison with the labeled dataset to verify the labeling. The survey images used for verifying the scene labeling are shown in Fig. 8. The middle of Fig. 8 is the composite image of the label image and the SAR image. The survey images with their locations are surrounding the middle figure. The survey images with very high resolution can ensure that the labeling processes and results are sound for model training, evaluation, and comparison.

E. Construction of the Dataset

In this study, four types of input information compositions are considered: 1) VV polarization; 2) VH polarization; 3) VV + VH dual polarization; and 4) VV + VH + W + T (W for wind speed and T for tide level) dual-polarization with wind speed and tide level information.

In the study area, the size of each image from one acquisition is 2192×2740. We randomly crop the initial images into image patches with the size of 256×256 to construct the dataset. A dataset with 20 076 image patches is constructed. We use data augmentation strategies, such as random flip and

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TABLE II
CLASSIFICATION RESULTS WITH DIFFERENT INPUT INFORMATION COMBINATIONS OF THE TESTING IMAGES

Dataset	Class	VV	VH	VV+VH	VV+VH+W+T
20161018	Water	99.65	98.99	**99.74**	99.63
	Sediment	95.49	54.84	**97.30**	96.59
	Dense Vegetation	78.94	59.53	82.65	**84.78**
	Sparse Vegetation	82.50	85.25	86.59	**88.77**
20170907	Water	99.36	99.14	**99.56**	98.37
	Sediment	90.38	42.68	89.34	**92.40**
	Dense Vegetation	67.34	60.82	**69.98**	67.52
	Sparse Vegetation	75.56	81.57	80.60	**83.07**
20181008	Water	97.14	**98.00**	97.20	97.11
	Sediment	95.60	35.78	96.98	**97.46**
	Dense Vegetation	**86.04**	83.31	75.99	84.63
	Sparse Vegetation	79.28	83.01	**88.05**	73.66
20191108	Water	99.42	96.14	**99.43**	99.38
	Sediment	64.62	72.47	68.24	**79.80**
	Dense Vegetation	90.25	**92.15**	90.71	88.74
	Sparse Vegetation	90.66	84.40	**94.00**	92.71
20200909	Water	98.18	99.23	99.08	**99.31**
	Sediment	**92.84**	43.60	86.90	90.55
	Dense Vegetation	73.12	78.76	87.36	**87.52**
	Sparse Vegetation	73.34	**76.03**	68.51	73.67
Overall	Water	98.80	98.23	**99.04**	98.80
	Sediment	92.55	45.36	92.12	**93.97**
	Dense Vegetation	78.70	79.41	**82.75**	81.54
	Sparse Vegetation	81.37	81.75	83.82	**84.76**
OA (%)	-	95.53	86.75	95.97	**96.15**
mIoU	-	0.7922	0.6592	0.8193	**0.8222**

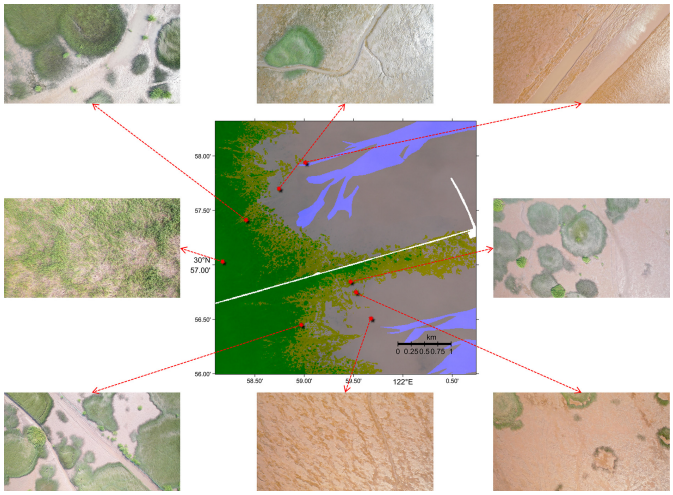


Fig. 8. Survey images of the Shanghai Nanhui intertidal zone on September 10, 2020. Middle image: composite image of the label image and the SAR image. Surrounding images: aerial images taken by a drone with their locations. In the middle image, we put the labeled image as a layer superimposed onto the SAR image via Photoshop Software. In this way, we can check the quality of the manual annotation.

rotation, to enhance the model's generalization ability. The image data are normalized into data with a mean of zero and a standard deviation of 0.5.

The dataset is divided into three subsets: the training set, the validation set, and the testing set. From the dataset, 85 images are selected to train the model. From 2016 to 2020, we select one image as the testing image for every year. We hope that, in each testing image, all the land cover types are sufficiently represented. In addition, all the years in the study interval are sufficiently represented. The ratio between the training set and the validation set is 3 to 1. The training set is used to learn the parameters of the models, and the validation set is used to determine which set of parameters is chosen.

V. RESULTS AND DISCUSSIONS

A. Experimental Settings and Assessment Criteria

For the experimental settings, the total training epoch is set to 300. After checking the validation loss, we find that this total training epoch can fit the models with different settings. The batch size is set to 16. The model optimizer is Adam [79], and the initial learning rate (LR) is set to 0.001. In the LR adjustment scheme adopted by the optimizer, for every 30 epochs of training, if the loss has no apparent downward trend, the LR will decrease to 0.2 times the current LR.

To evaluate how well the models perform on the experimental data, two indicators are selected, overall accuracy (OA) and mean intersection over union (mIoU). OA represents the percentage of the model's correctly labeled pixels. IoU is the ratio of the intersection to the union of the two sets, prediction and ground truth, of each category. The IoU of each category is calculated, accumulated, and then averaged to get the mIoU.

B. Classification Results on Single-Polarization Input Information

In this part, U^2-Net is used to train the SAR dataset with single-polarization input information, and the trained models are evaluated on the testing data of five different dates. The classification results are shown in Table II.

On the testing images, the model with VH polarization input information achieves the OA of 86.75% and the mIoU of

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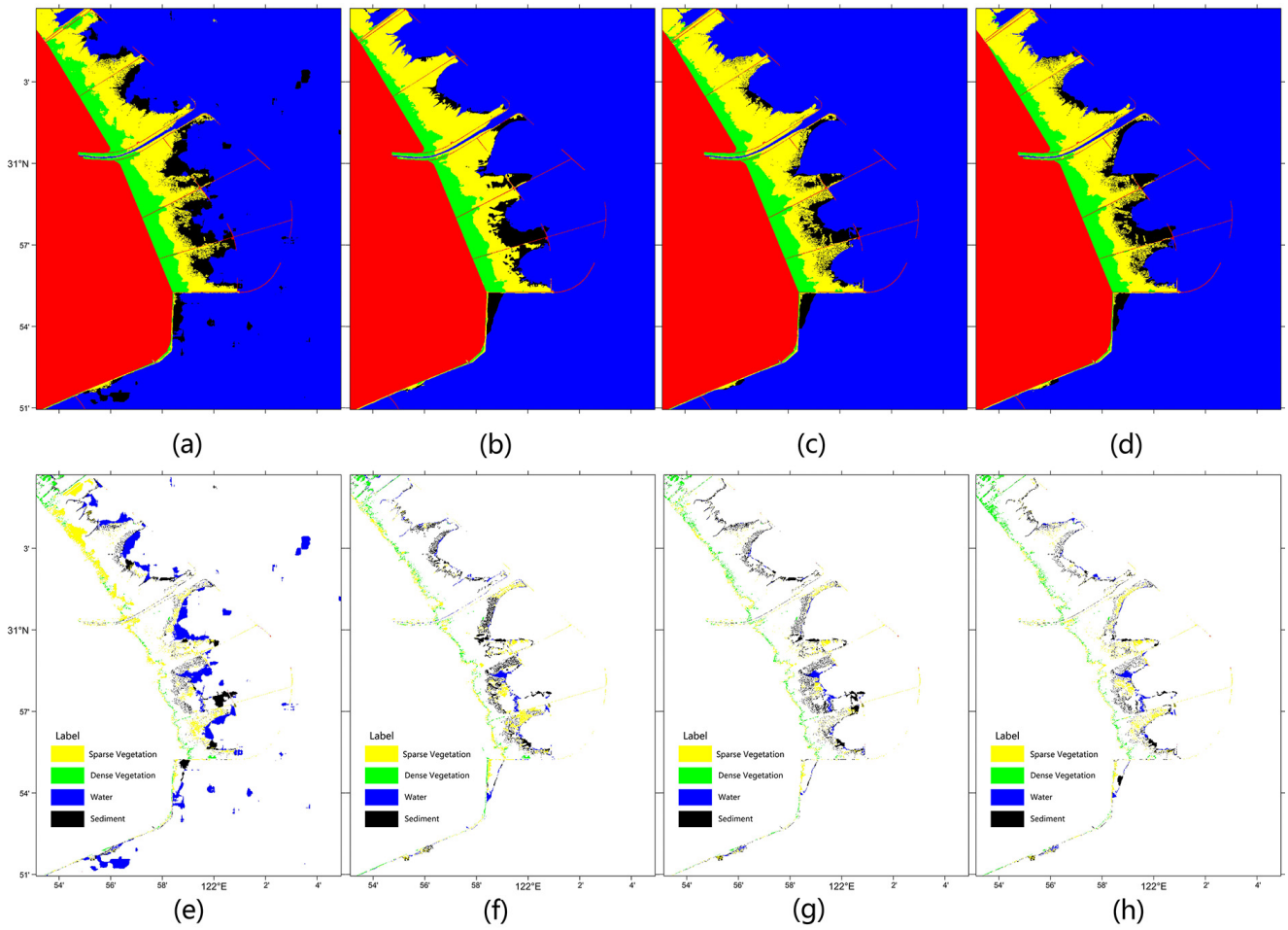


Fig. 9. Classification maps using different input information combinations of the testing image on November 8, 2019. First row: classification map. Second row: misclassified pixels in the label image. (a) and (e) VH. (b) and (f) VV. (c) and (g) VV + VH. (d) and (h) VV + VH + W + T.

0.6592. The model with VV polarization input information achieves the OA of 95.53% and the mIoU of 0.7922, and it is better than the model trained with VH polarization input information. These experimental results show that in the case that only single-polarization information can be obtained, the VV polarization channel of SAR data is more suitable for the intertidal zone's classification task than the VH polarization channel. In terms of specific categories, incorrectly classified categories on the VH polarization input information are mainly concentrated in water and sediment. The possible reasons are: first, in the VH polarization channel, the radar backscattering characteristics of sediment and water are similar; second, the VH noise floor is high. The VV polarization input information classification has better prediction performance in sediment. The possible reason is that the VV polarization channel is sensitive to surface roughness. Most of the sediment in the study site is formed after tide level drops, and its surface is much smoother than the water surface. The scattering type of sediment is mainly specular scattering. Therefore, it is easier to use the VV polarization channel to distinguish the scattering of sediment from the scattering of water. For vegetation with two different coverage types, misclassified categories in the VV and VH polarization channels are mainly concentrated in the two types. The radar backscattering characteristics between vegetation coverage areas are similar, with no apparent boundaries. The above-mentioned analyses show that the classification of single-polarization input information still has room for performance improvement in the intertidal zone land cover classification.

Taking the image of November 8, 2019, as an example, the classification results are shown in Fig. 9. According to overall classification performance visual effects, the classification of VV single-polarization input information is better than VH classification.

C. Classification Results on Multisource Input Information

In this part, U^2-Net is used to train the SAR dataset with the input information of VV + VH and VV + VH + W + T, and the trained models are evaluated on the testing data of five different dates. The classification results are shown in Table II.

Table II shows the classification performance on VV + VH and VV + VH + W + T input information. The OA, mIoU, and individual classification accuracies are improved. Specifically, the classification on VV + VH achieves the OA of 95.97% and the mIoU of 0.8193. The classification on VV + VH + W + T achieves the OA of 96.15% and the

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TABLE III
CLASSIFICATION RESULTS WITH DIFFERENT MODELS OF THE TESTING IMAGES (THE PROPOSED MODEL VERSUS THE BENCHMARKING ONES)

Dataset	Class	SVM	U-Net	U^2-Net	MB-U^2-ACNet
20161018	Water	74.18	**99.72**	99.63	99.68
	Sediment	57.87	96.52	96.59	**97.54**
	Dense Vegetation	**87.01**	85.69	84.78	76.51
	Sparse Vegetation	51.40	81.95	88.77	**90.19**
20170907	Water	66.56	98.51	97.11	**99.02**
	Sediment	66.17	93.95	**97.46**	92.12
	Dense Vegetation	62.85	70.16	**73.66**	65.97
	Sparse Vegetation	55.14	78.46	**84.63**	81.47
20181008	Water	75.15	97.31	**98.37**	97.18
	Sediment	62.19	95.83	92.40	**97.61**
	Dense Vegetation	84.52	76.63	67.52	**86.42**
	Sparse Vegetation	55.96	**85.02**	83.07	76.87
20191108	Water	79.07	99.43	99.38	**99.46**
	Sediment	40.97	66.12	**79.80**	78.96
	Dense Vegetation	86.93	88.01	88.74	**91.84**
	Sparse Vegetation	57.78	92.05	**92.71**	91.70
20200909	Water	75.48	99.22	**99.31**	98.65
	Sediment	61.89	88.77	90.55	**96.23**
	Dense Vegetation	69.68	**87.86**	87.52	87.50
	Sparse Vegetation	56.44	69.88	**73.67**	72.65
Overall	Water	74.21	**98.87**	98.80	98.84
	Sediment	61.19	93.05	93.97	**95.24**
	Dense Vegetation	75.73	82.61	81.54	**84.60**
	Sparse Vegetation	56.02	82.20	**84.76**	82.87
OA (%)	-	70.62	95.90	96.15	**96.40**
mIoU	-	0.4664	0.8147	0.8222	**0.8307**

mIoU of 0.8222. The results are 0.0300 for mIoU higher than the classification results on VV.

In Fig. 9, the classification results on VV + VH and VV + VH + W + T input information show good performance according to visual inspection. Every type of predicted land cover is close to the labeled ground truth. The misclassified pixels are primarily concentrated in the category boundaries since the boundaries between classes in SAR images are not clearly shown. Another possible reason is that the information from different sources can be more effectively integrated. We will show this in the following part.

By means of the above-mentioned two experiments, it can be concluded that the classification of the dual-polarization input information achieves better performance than that of the single-polarization input information in both overall and individual classification performances. Furthermore, integrating wind and tide level information into the framework achieves even better performance. This indirectly answers the question of improving the DCNN-based model's performance for flood mapping under the influence of strong wind, raised by Liu *et al.* [46].

D. Comparison With the Benchmarking Methods

In this part, we compare the MB-U^2-ACNet model with three benchmarking methods, U^2-Net, U-Net [67], and SVM [80], to show the advantages of the proposed model. It is worth noting that after experiment verification on the dataset, VV as branch 1 with VH + W + T as branch 2 is the channel combination with better performance. Therefore, this combination is used in the MB-U^2-ACNet model.

The U-Net model is a U-shaped structure with the encoding and decoding parts. This network is one of the most widely used network structures in the deep learning-based image semantic segmentation field. It has been extended from the original medical image segmentation task to crop mapping [81], sea–land segmentation [82], shadow recognition [83], and other applications in the field of remote sensing. The U-Net model in this study uses the same training strategy listed earlier and input data cube for input information fusion.

The SVM has been used to classify features in remote sensing for a long time. It is a type of representative method in the shallow machine learning era. This study uses the SVM method to classify the integrated input information as features in the intertidal zone. The main parameters for the SVM model are listed as follows: the radial basis function as the kernel function; the parameter gamma is set to 0.01; and the penalty coefficient is set to 100.

The above-mentioned methods are all trained and tested on the same training and testing subsets to ensure objectivity. The performance comparison is listed in Table III, and the results on the image of November 8, 2019, are shown in Fig. 10.

From Table III, the proposed MB-U^2-ACNet model is obviously better than the benchmarking SVM and U-Net models in the intertidal zone land cover classification using dual-polarimetric SAR data integrated with environmental information. The mIoU of the MB-U^2-ACNet is 0.0160 and 0.3643 higher than the mIoUs of the U-Net and the SVM, respectively. The main reason is that the proposed model enhances its overall complexity by adding more substructures to extract richer global and local information in the intertidal

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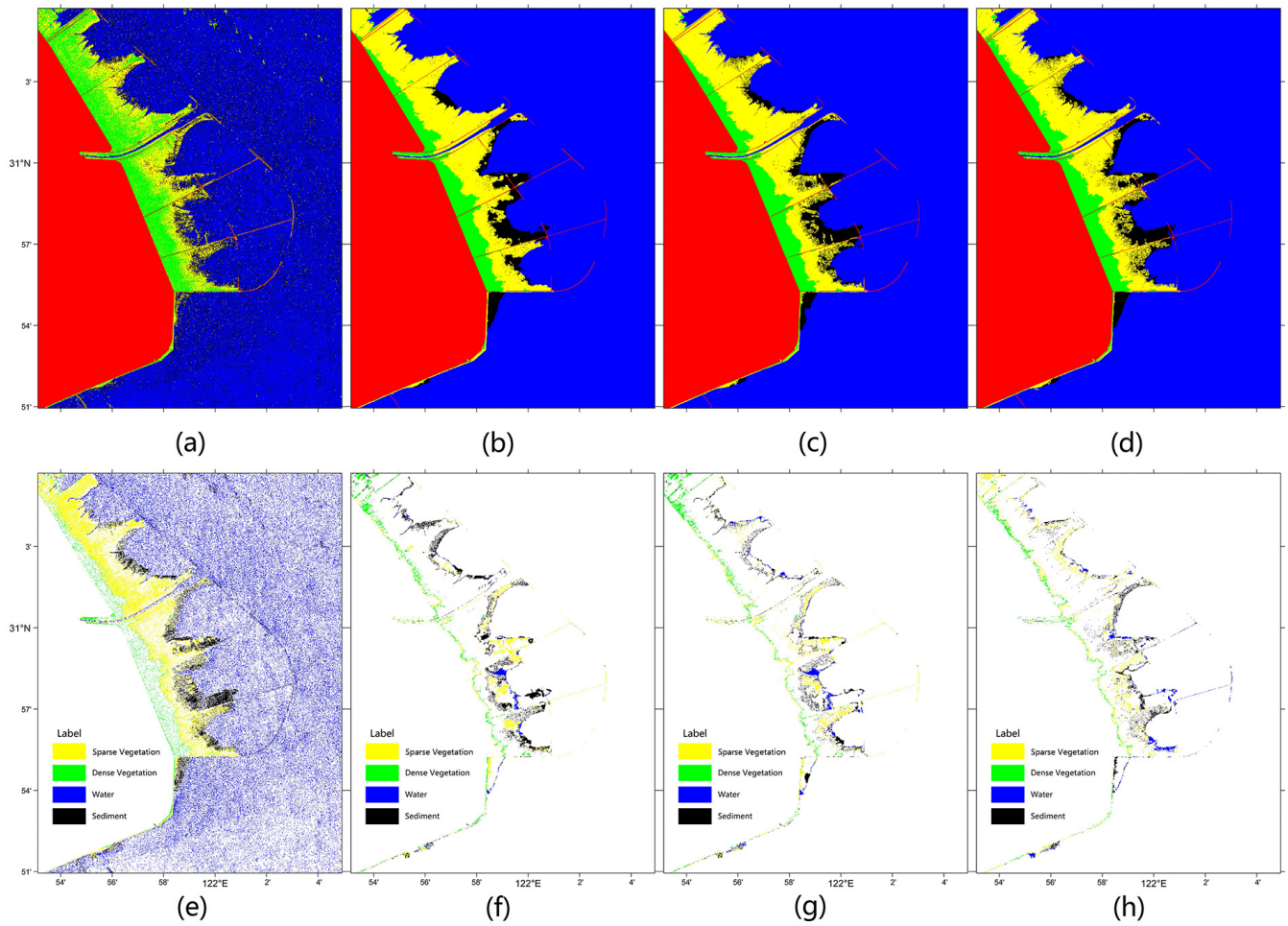


Fig. 10. Classification maps using different models (the proposed one versus the benchmarking ones) of the testing image on November 8, 2019. First row: classification map. Second row: misclassified pixels in the label image. (a) and (e) SVM. (b) and (f) U-Net. (c) and (g) U^2-Net. (d) and (h) MB-U^2-ACNet.

zone and learn more complex and abstracted high-level semantic features.

MB-U^2-ACNet performs better than U^2-Net, due to the contributions of the ACRSU, the attention modules, and the multibranch feature fusion structure. We will cover this in the following two parts.

E. Substructure Analysis

In this part, we perform the substructure analyses for the proposed model.

1) Effectiveness of ACRSU:

This study compares the classification results between U^2-Net with ACRSU (U^2-ACNet) and without ACRSU using the random-flip-and-rotation data augmentation strategies. In addition, the classification results of U^2-Net with ACRSU but without data augmentation (U^2-ACNet-NDA, NDA stands for no data augmentation) and U^2-Net without ACRSU nor data augmentation (U^2-Net-NDA) are included for comparison. Table IV shows the OAs and mIoUs of the comparing experimental results.

One can observe that the performance of the U^2-ACNet is better than that of the U^2-Net, showing that the ACRSU module can improve the model's analytical ability and then

TABLE IV
CLASSIFICATION RESULTS WITH DIFFERENT U-BLOCKS OF THE TESTING IMAGES

	U^2-Net-NDA	U^2-Net	U^2-ACNet-NDA	U^2-ACNet
OA(%)	96.07	96.15	96.23	**96.31**
mIoU	0.8211	0.8222	0.8246	**0.8279**

contribute to classification performance improvement. Furthermore, the performance of the U^2-ACNet is better than that of the U^2-ACNet-NDA. This result shows that data augmentation contributes to performance improvement.

2) Effectiveness of Attention Modules:

Here, we test the effectiveness of the attention modules. In the proposed model, the spatial attention and channel attention modules are used. The spatial attention module is connected after each ACRSU-block operation in the branching structure and before the feature fusion operation. The channel attention module is connected after the feature maps outputted by each decoding stage. This design's basis was introduced in Section II. To present the contribution of the attention modules, we conduct the following experimental design. First, we only add the spatial attention modules to the MB-U^2-Net with the multibranch structure, and we name it MB-U^{2}Net-S. Second, we only add the channel attention modules to the MB-U^2-Net, namely, MB-U^{2}Net-C. Finally, both the spatial and channel

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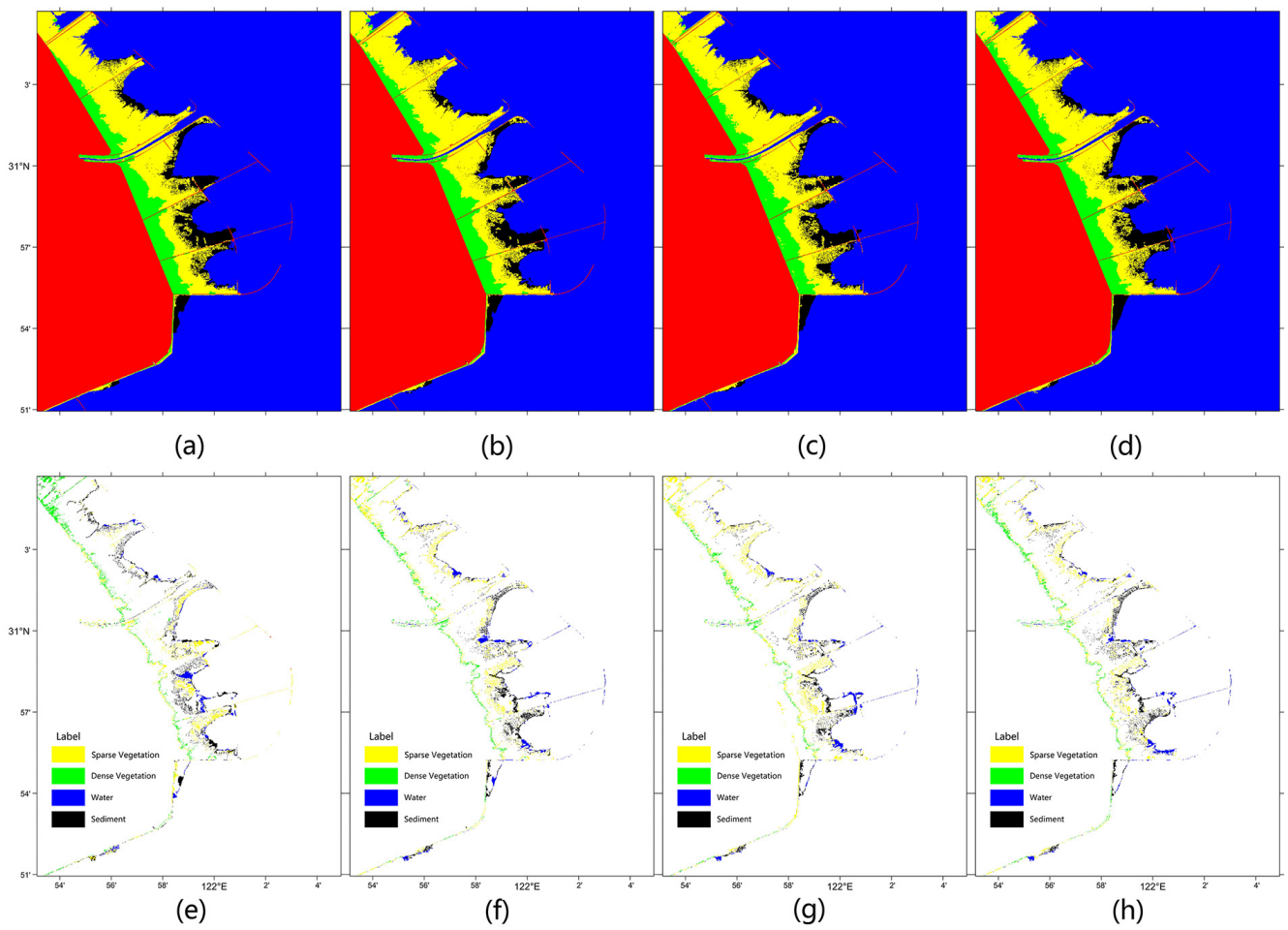


Fig. 11. Classification maps using different models (the proposed one versus the ablated ones) of the testing image on November 8, 2019. First row: classification map. Second row: misclassified pixels in the label image. (a) and (e) U^2-Net. (b) and (f) U^2-Net-V1. (c) and (g) U^2-Net-V2. (d) and (h) MB-U^2-ACNet.

TABLE V
CLASSIFICATION RESULTS WITH DIFFERENT ATTENTION MODULES OF THE TESTING IMAGES

	MB-U^2-Net	MB-U^{2}Net-C	MB-U^{2}Net-S	MB-U^{2}Net-SC
OA(%)	96.24	96.28	96.33	**96.39**
mIoU	0.8252	0.8274	0.8280	**0.8290**

attention modules are added to the MB-U^2-Net, and we named it MB-U^{2}Net-SC. Table V lists the OAs and mIoUs of the models.

Experimental results show that the two attention module addition schemes designed in our model can capture important spatial and channel information in the feature maps, enhance feature recognition capabilities, and improve classification accuracy.

3) Effectiveness of the Multibranch Feature Fusion Structure

From the last part, we can observe that the performance of MB-U^2-ACNet is better than that of U^2-Net, due to the contributions of the ACRSU, the attention modules, and the multibranch feature fusion structure. Table IV shows the contribution of the ACRSU design. Table V shows the contribution of the attention modules. Comparing the result of the U^2-Net in Table III and the result of the MB-U^2-Net in Table V, we can observe that the latter is better than the former. This result verifies the contribution of the structure of multibranch feature fusion. This structure can introduce a more heterogeneous and flexible way for deep feature fusion and further improve classification performance.

F. Ablation Study

In this part, we continue to analyze the contributions of each substructure more systematically by the ablation study. Between the U^2-Net and the proposed MB-U^2-ACNet, three substructures are constructed. We can add the substructures one by one. First, we construct the U^2-Net-V1, and this model adds a multibranch structure based on the initial U^2-Net model. Based on the U^2-Net-V1, we construct the U^2-Net-V2, and this model adds attention modules to the backend of the RSU blocks and the decoding sides. The U^2-Net-V1 is the same as the MB-U^2-Net with the multibranch structure of the last part, and the U^2-Net-V2 is the same as the MB-U^{2}Net-SC with the multibranch structure and the attention modules. The experimental results are listed in Table VI. For the convenience of visual comparison, the results on the image of November 8, 2019, are shown in Fig. 11.

From Table VI, we can see that the proposed model shows the best results. From the U^2-Net to the MB-U^2-ACNet, the

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TABLE VI
CLASSIFICATION RESULTS WITH DIFFERENT MODELS OF THE TESTING IMAGES (THE PROPOSED MODEL VERSUS THE ABLATED ONES)

Dataset	Class	U^2-Net	U^2-Net-V1	U^2-Net-V2	MB-U^2-ACNet
20161018	Water	99.63	99.56	99.66	**99.68**
	Sediment	96.59	96.07	97.13	**97.54**
	Dense Vegetation	**84.78**	84.14	83.59	76.51
	Sparse Vegetation	88.77	**90.21**	91.42	90.19
20170907	Water	97.11	98.95	98.68	**99.02**
	Sediment	**97.46**	92.00	93.59	92.12
	Dense Vegetation	**73.66**	67.70	66.21	65.97
	Sparse Vegetation	**84.63**	81.78	81.53	81.47
20181008	Water	**98.37**	97.16	97.23	97.18
	Sediment	92.40	96.80	**97.73**	97.61
	Dense Vegetation	67.52	81.73	82.30	**86.42**
	Sparse Vegetation	83.07	**83.76**	81.30	76.87
20191108	Water	99.38	99.46	**99.55**	99.46
	Sediment	**79.80**	76.03	75.05	78.96
	Dense Vegetation	88.74	91.09	88.45	**91.84**
	Sparse Vegetation	92.71	92.17	**92.95**	91.70
20200909	Water	**99.31**	98.74	98.87	98.65
	Sediment	90.55	93.70	94.97	**96.23**
	Dense Vegetation	87.52	87.51	**87.52**	87.50
	Sparse Vegetation	**73.67**	73.12	71.41	72.65
Overall	Water	98.80	98.82	98.84	**98.84**
	Sediment	93.97	94.02	95.14	**95.24**
	Dense Vegetation	81.54	83.82	83.15	**84.60**
	Sparse Vegetation	**84.76**	84.30	83.83	82.87
OA (%)	-	96.15	96.24	96.39	**96.40**
mIoU	-	0.8222	0.8252	0.8290	**0.8307**

models' classification performances are gradually improved due to the addition of different architectures. The addition of the multibranch structure from the U^2-Net to the U^2-Net-V1 improves the model's ability to analyze and fuse information from different sources. Compared with the benchmarking U^2-Net model, the U^2-Net-V1 improves the mIoU by 0.0030. The addition of the spatial and channel attention modules from the U^2-Net-V1 to the U^2-Net-V2 improves the model's ability to focus on salient features in the spatial and channel dimensions. Compared with the U^2-Net-V1, the U^2-Net-V2 further improves the mIoU by 0.0038. From the U^2-Net-V2 to the MB-U^2-ACNet, the proposed ACRSU blocks are used to replace the original RSU blocks, enhancing the convolutions for feature extraction. Compared with the U^2-Net-V2, the MB-U^2-ACNet further improves the mIoU by 0.0017. Compared with the benchmarking and sophisticated U^2-Net model, the proposed MB-U^2-ACNet model improves the mIoU by 0.0085. Here, by ablation study, the effectiveness of the proposed MB-U^2-ACNet model in the intertidal zone environment monitoring using dual-polarimetric SAR data integrated with environmental information is verified systematically.

G. Impact of the Proportion of Training Samples

This part will analyze the impact of the proportion of training samples. Since the performance of the SVM model is significantly worse than those of others, the SVM model is not included for comparison in this part. Instead, we select the MB-U^2-ACNet, U^2-Net-V2, U^2-Net-V1, U^2-Net, and U-Net models for comparison. From the training subset, from 20% to 100%, training samples are randomly selected (not exactly randomly for 100%). For different proportions of training samples, we adopt the same hyperparameters and training strategies to ensure the results' reliability. Fig. 12 shows that the classification accuracies increase as the volumes of training samples increase. The experimental results confirm the advantages of the proposed MB-U^2-ACNet model under the influence of the proportion of training samples. As the proportion of training samples decreases from 1.0 to 0.2, the performance of every model decreases. This is in line with our understanding. However, the MB-U^2-ACNet has the gentlest decline. From another perspective, this shows the performance of the MB-U^2-ACNet is more advantageous and stable.

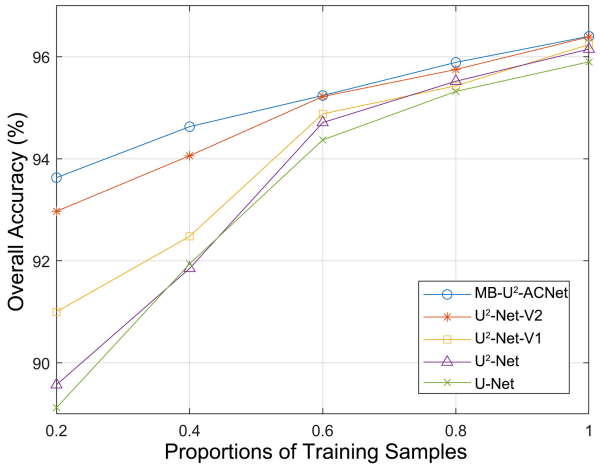


Fig. 12. Classification results with different proportions of training samples.

VI. CONCLUSION

This article proposes an MB-U^2-ACNet model suitable for intertidal zone land cover classification using

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dual-polarimetric SAR data integrated with environmental information. The MB-U^2-ACNet model is specially tailored from a sophisticated deep learning-based semantic segmentation model U^2-Net. The MB-U^2-ACNet model takes dual-polarimetric SAR data and environmental information as input and uses a multibranch feature fusion structure to extract features from different channels. Then the branching structure information is merged through concatenation. Each stage of the MB-U^2-ACNet is filled by the designed ACRSU blocks based on the asymmetric convolution. Therefore, it can improve the model's ability to extract local and global features. Furthermore, the model adds attention modules to the output end of each branch unit and the decoders' sides so that the network can focus on more discriminative channel and spatial features. The Shanghai Nanhui intertidal zone is selected as a studying case. In the dataset constructed for the Shanghai Nanhui intertidal zone environment monitoring, the proposed MB-U^2-ACNet model demonstrates its advantages, compared with the benchmarking methods. The model reaches the classification accuracy of 96.40% and the mIoU score of 0.8307. The experiments on the dataset show the advantages of the proposed model in intertidal zone environment monitoring due to better feature extraction and multisource information fusion. Moreover, the contribution of every added substructure is analyzed systematically. In this article, we use Shanghai Nanhui intertidal zone as a study case. Actually, we treat this case as a general one, without using specific information other than remote sensing data and environmental information. We believe the proposed model is ready to be generalized to other intertidal zone areas for environment monitoring. In order to further improve the classification performance, we could treat input time series as a whole instead of single-time imagery. However, the cost is the model could be much more complicated.

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